Abhyudaya Yadav

Roll No.:2511AI04
Master of Technology
Mathematics and Computing
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech. (AI)	Indian Institute of Technology, Patna	67.6 %	2025-2027
B.Tech. (Hons.) (CSE)	Shri Shankaracharya Technical Campus, Bhilai	85.9 %	2019-2023
Senior Secondary	Swami Vivekanand Senior Secondary School	63.8 %	2017
Secondary	Holy Hearts Educational Academy	85.5 %	2015

Experience

- **Graduate Student Researcher** *Jun. 2026 - Present*
  Cybersecurity Lab,IIT Patna Patna, Bihar
  - Developing an AI-driven insulin pump security digital twin that couples SimGlucose-based physiological simulation with ML/DL glucose prediction and GECO-based physiological modeling to establish patient-specific safety expectations.
  - Integrating hazard and survival analysis with reachability-based safety constraints to assess emerging risks, while cyberattack detection and evidence fusion inform risk-aware decisions for insulin delivery monitoring and safety responses.
  - Building an NVIDIA Omniverse/OpenUSD interface to visualize patient physiology, pump behavior, twin predictions, and security decisions in real time.

- **Research & Development Intern** *Jul. 2025 - Jun 2026*
  CVPR Lab, IIT Patna Patna, Bihar
  - Conducted research focusing on temporal recommender systems, evolving user preferences, GloVe-based semantic representations, behavioral drift analysis, and predictive modeling using Random Forest, PCA, K-Means, and DBSCAN.
  - Developed TrustShield-VLM, an OpenCLIP-based semantic integrity framework, analyzing 156 image-text samples through 512-dimensional embeddings and a 1,560-edge trust graph, integrating six indicators to assess multimodal representation reliability.

- **Teaching Assistant** *JuL. 2025 - Present*
  IIT Patna Patna, Bihar
  - Served as a Teaching Assistant for CS1201 Data Structures Lab, CS1101 Foundations of Programming Lab, CS2104 Discrete Mathematics at IIT Patna.
  - Assisted in conducting lab sessions and tutorials while mentoring undergraduate students in data. structures, programming fundamentals, algorithmic thinking, debugging techniques, and structured problemsolving approaches.

Projects

- **EduNova SLM AI-Powered Teaching Assistant** *Feb. 2026*
  IIT Patna Patna, Bihar
  - EduNova SLM is a Transformer-powered Teaching Assistant AI that combines Hugging Face models, Sentence Transformers, and Rank-BM25 for hybrid information retrieval. It generates conceptual explanations, numerical solutions, and exam-ready questions using lightweight SLM reasoning, enabling accurate, syllabus-aligned educational assistance with efficient, scalable NLP pipelines.

- **Hybrid Adaptive Cross-Users Movie Recommender System** *Dec. 2025 - Jun 2026*
  IIT Patna Patna, Bihar
  - Developed a Python-based recommendation framework for analyzing large-scale user interaction datasets through temporal preference modeling. Designed data processing and experimentation pipelines for feature extraction, behavioral analysis, and predictive evaluation using statistical techniques, enabling scalable analysis of evolving user preferences and data-driven decision-making.

Positions of Responsibility

- **Member** AMD & NVIDIA Developer Community *Sept. 2026 - Present*
- **Placement Coordinator** at Centre for Career Development and Counselling, IIT Patna *Jan. 2026 - Present*
- **Member,** Google Developer Groups, SSTC, Bhilai *Jul. 2020 - May 2023*
- **Campus Ambassador,** Techfest, IIT Bombay *Jul. 2020 - Feb 2021*

Technical Skills

–**Programming & Development:** Python, C++, C, Bash, Git, GitHub, SQL, MySQL

–**Machine Learning & Deep Learning:** PyTorch, Scikit-learn, Transformers, Neural Networks, Classification, Regression, Clustering, Feature Engineering, PCA, Random Forest

–**LLMs, NLP & Generative AI:** Large Language Models (LLMs), Small Language Models (SLMs), Natural Language Processing, RAG, Prompt Engineering, Hugging Face, Embeddings, AI Agents

–**Recommender Systems & Data Analytics:** Recommender Systems, User Behavior Modeling, Temporal Modeling, Preference Drift Analysis, Time-Aware Recommendation, NumPy, Pandas, Matplotlib, Statistical Analysis, Data Visualization, BM25, FAISS

–**AI Security:** Digital Twins, Anomaly Detection, Residual Analysis, CUSUM, Data Poisoning, Adversarial Analysis, NVIDIA Omniverse

Achievements

– **Received Honours in Bachelor Degree** for securing 8.59 CGPA *2023*
– **Google Kick Start 2021 Round E** Rank 3530th *2021*
– **Secured NPTEL Elite** Problem Solving through Programming in C, MHRD, Govt. of India. Secured 60%. *2021*
– **CodeChef Maximum Rating: 1437** Qualified for CodeChef SnackDown 2021 Round A *2021*
– **Qualified JEE Mains 2019** by securing 80.59 percentile. *2019*

Areas of Interests

* Machine Learning
* Natural Language Processing
* Recommender System
* Deep Learning
* Generative AI
* Multimodal AI
* Applied AI Research

Co-Curricular Activities

* 6k+ followers on LinkedIn, 450+ followers on Instagram
* Owns and Manages YouTube Channel "Travelman Vlogs" with 100+ subscribers based on my travel experiences.
* Achieved Google Local Guide Level 5.
* Won quarter-final in Table Tennis tournament held at SSTC, Bhilai.
* Participated in SSTC, Bhilai photography and quiz contest.

Certifications

* **NPTEL Elite** in Problem Solving through programming in C
* **Deep Learning.AI** Neural Networks & Deep Learning
* **Bolt-IOT** Machine Learning Training
* **Cisco Networking Academy** Javascript Essentials I
* **Cisco Networking Academy** Networking Basics

Languages

* English
* Hindi